

JUN YUAN

Postdoctoral Research Fellow, University of Virginia

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My research lies at the intersection of **interpretable AI**, **data visualization**, and **human-AI systems**. I study how people can understand, interact with, and retain meaningful agency over AI systems, from explaining individual algorithmic decisions to designing human-AI systems that remain **interpretable, robust, and accountable over time**. My work combines machine learning, interactive system design, and human-centered evaluation.

ACADEMIC APPOINTMENT

University of Virginia

Postdoctoral Research Fellow, Digital Technology for Democracy Lab

Charlottesville, VA

Aug. 2025 – Present

Mentor: Steven L. Johnson

EDUCATION

New Jersey Institute of Technology

Ph.D. in Data Science

Newark, NJ

Jan. 2019 – Dec. 2024

Advisor: Aritra Dasgupta

Dissertation: Visual Analytic Techniques for Interpretable Algorithmic Ranking Systems

Clemson University

M.S. in Mathematical Sciences

Clemson, SC

Aug. 2015 – Aug. 2018

Emphasis on Statistical Modeling.

Hangzhou Dianzi University

B.S. in Mathematics and Applied Mathematics

Hangzhou, China

Sept. 2010 – Sept. 2014

Concentration in Scientific Computing.

PUBLICATIONS

11. **J. Yuan**. 2026. Interchangeable Agency: Anticipated Replacement Creates Legibility Pressure in Human-AI Handoffs. *AAAI Fall Symposium on Trustworthy Agentic Systems (TAS 2026)*. (Accepted)
10. **J. Yuan**, X. Ye. 2026. The Cross-Context Threshold Test: Detecting Discrimination Under Environmental Shifts. *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)*, PMLR 300:4330–4338. [Published](#).
9. **HLE Contributors Consortium** (including **J. Yuan**). 2026. A Benchmark of Expert-Level Academic Questions to Assess AI Capabilities. *Nature*, 649, 1139–1146. (Expert question contributor) [Published](#).
8. K. Bhattacharjee, **J. Yuan**, A. Dasgupta. 2025. ReVise: A Human-AI Interface for Incremental Algorithmic Recourse. *2025 IEEE Visualization and Visual Analytics (VIS)*. [Published](#).
7. C. Fitzpatrick, **J. Yuan**, A. Dasgupta. 2024. TalkToRanker: A Conversational Interface for Ranking-based Decision-Making. *VISxAI Workshop at IEEE VIS 2024*. [Published](#).
6. V. Koli, **J. Yuan**, A. Dasgupta. 2024. Sensemaking of Socially-Mediated Crisis Information. *Proceedings of the Third Workshop on Bridging Human-Computer Interaction and Natural Language Processing*. [Published](#).
5. **J. Yuan**, K. Bhattacharjee, A. Z. Islam, A. Dasgupta. 2024. TRIVEA: Transparent Ranking Interpretation using Visual Explanation of Black-Box Algorithmic Rankers. *The Visual Computer*, 40, 3615–3631. [Published](#).
4. **J. Yuan**, A. Dasgupta. 2023. A Simple Scoring Function to Fool SHAP: Stealing from the One Above. *XAI in Action Workshop at NeurIPS 2023*. [Published](#).

3. M. Sloane, I. R. Solano-Kamaiko, **J. Yuan**, A. Dasgupta, J. Stoyanovich. 2023. Introducing Contextual Transparency for Automated Decision Systems. *Nature Machine Intelligence*. [Published](#).
2. **J. Yuan**, A. Dasgupta. 2023. A Human-in-the-loop Workflow for Multi-Factorial Sensitivity Analysis of Algorithmic Rankers. *Human-in-the-Loop Data Analytics Workshop at SIGMOD 2023*. [Published](#).
1. **J. Yuan**, J. Stoyanovich, A. Dasgupta. 2022. Rankers, Rankees, & Rankings: Peeking into the Pandora's Box from a Socio-Technical Perspective. *Interrogating Human-Centered Data Science Workshop at CHI 2022*. [Published](#).

MANUSCRIPTS IN PREPARATION

- **J. Yuan**, A. Dasgupta. 2027. Fooling SHAP with Output Shuffling Attacks.
- V. Koli, B. Divaaniaazar, **J. Yuan**, Y. Meng, E. Deess, A. Dasgupta. 2027. ConsensusLens: A Design Study for Interpreting Course Evaluation Student Surveys.
- **J. Yuan**. 2027. A Scalable Human-AI Annotation Pipeline for Low-Resource Spoken Languages.

RESEARCH EXPERIENCE

University of Virginia

Charlottesville, VA

Postdoctoral Research Fellow

Aug. 2025 – Present

- Study long-horizon human-AI systems in which responsibility and task state pass between human and AI agents; develop methods for evaluating interpretability, continuity, robustness, and meaningful human agency during handoffs.
- Developed a human-AI annotation pipeline for a low-resource Wenzhounese speech corpus, studying audio-text segment alignment and how annotators' language proficiency and value preferences shape annotation practices.
- Contributed to grant proposal development for *EveryTown EveryShot*, extending prior work to study how media coverage of gun violence may shape police deployment patterns.

New Jersey Institute of Technology

Newark, NJ

Research Assistant

Jan. 2019 – Dec. 2024

- Developed interpretable and interactive ranking systems, including TRIVEA, CARE, Recourse, HILSA, and TalkToRanker, to support understanding and intervention in algorithmic decision-making.
- Studied vulnerabilities of explainable AI methods and developed adversarial strategies for manipulating feature-attribution explanations such as SHAP.
- Collaborated with university administrators, social scientists, responsible AI researchers, and industry partners on applications including student admissions, course evaluation, and AI-assisted hiring.

Accern Inc.

New York, NY

Data Science Intern

Jun. 2021 – Sept. 2021

- Developed BERT-based subjectivity models for COVID-19 social-media data and evaluated NLP annotation workflows for financial decision-support applications.

Accern Inc.

New York, NY

Data Science Intern

Jun. 2020 – Sept. 2020

- Applied BERT-based topic modeling and t-SNE to financial news to identify temporal topic trends during the COVID-19 outbreak.

New Jersey Institute of Technology

Newark, NJ

Research Mentor, NSF REU Summer Research Program

Jun. 2019 – Sept. 2019

- Mentored three undergraduate researchers on spatiotemporal analysis of racial disparities in NYPD Stop, Question, and Frisk data and led research meetings throughout the 10-week program.

TEACHING EXPERIENCE

New Jersey Institute of Technology

Newark, NJ

CS/DS 650: Data Visualization, Teaching Assistant

Jan. 2020 – Dec. 2023

- Mentored student project teams, reviewed course materials, and prepared homework and examination solutions for undergraduate and graduate students.

Clemson University

Clemson, SC

MATH 1010: Essential Mathematics for the Informed Society, Instructor of Record Sept. 2017 – Aug. 2018

- Independently taught a course of 40+ students, delivering two lectures per week. Coordinated grading, scheduling, and assessment development with instructors teaching other sections.

Clemson University

Clemson, SC

STAT 8010: Statistical Methods I, Lab Instructor

Jun. 2017 – Aug. 2017

- Taught SAS-based statistical analysis to graduate students in biology and chemistry; developed lab materials, guided analyses, and evaluated reports.

PRESENTATIONS & TALKS

- **J. Yuan.** “A Simple Scoring Function to Fool SHAP: Stealing from the One Above.” *XAI in Action Workshop at NeurIPS*, 2023.
- **J. Yuan.** “A Human-in-the-loop Workflow for Multi-Factorial Sensitivity Analysis of Algorithmic Rankers.” *Human-in-the-Loop Data Analytics Workshop at SIGMOD*, 2023.
- **J. Yuan.** “Ranking AI Visualization for Transparent Decision-making.” *Asian American Engineer of the Year Award Annual Convention*, 2023.
- **J. Yuan.** “Ranking AI Visualization for Transparent Decision-making.” *Chinese Association for Science and Technology, Greater New York Chapter (CAST-GNY)*, 2023.

AWARDS & HONORS

- 3rd Place, NJIT Research Day Competition, 2023.
- Graduate Student Organization of the Year, NJIT Graduate Student Association, 2020.

ACADEMIC SERVICE & LEADERSHIP

- Reviewer, *Towards Safe & Trustworthy Agents Workshop, NeurIPS 2024* Oct. 2024 – Nov. 2024
- Vice President of Finance, NJIT Graduate Student Association Sept. 2020 – Sept. 2021
- Vice President of Programming, NJIT Graduate Student Association Sept. 2019 – Sept. 2020

SKILLS

- **Research Areas:** Human–AI Interaction, Interpretable AI, Data Visualization, AI Agents, Algorithmic Ranking, Responsible AI, Human-Centered AI.
- **Methods & Tools:** Python, PyTorch, large language models, statistical modeling, learning-to-rank, interactive visualization, human-centered evaluation, Tableau, MATLAB, SAS, L^AT_EX.
- **Languages:** English, Mandarin Chinese, Wenzhounese.